NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Spine

1. cervical trauma
2. thoracolumbar trauma
3. degenerative cervical myelopathy
4. lumbar disc and stenosis
5. scoliosis and deformity
6. spinal tuberculosis and infection
7. spinal tumours and metastases
8. spinopelvic alignment

Cervical trauma

1. Clearance and imaging

- **NEXUS criteria** (no imaging needed if all five absent): **midline cervical tenderness, focal neurological deficit, altered alertness, intoxication, painful distracting injury**.
- **Canadian C-spine rule:** high-risk factors (**age 65 or more, dangerous mechanism, paraesthesia in the limbs**) = image; low-risk factors allowing range-of-motion assessment; able to rotate the neck 45 degrees left and right = no imaging. More sensitive and specific than NEXUS.
- **CT is the first-line imaging** in adult trauma (radiographs miss injuries, especially at the cervicothoracic junction and craniocervical junction). **MRI** for neurological deficit, suspected ligament/disc injury (**before reducing a facet dislocation in an obtunded patient**), SCIWORA, and for obtunded patients (debated: CT alone is sufficient to clear many obtunded patients with a high-quality negative CT).
- **Normal values: atlantodens interval (ADI) up to 3 mm in adults (5 mm in children)**; prevertebral soft tissue **under 6-7 mm at C2** and **under 22 mm at C6**; **Powers ratio** (BC/OA) under 1 (over 1 = anterior atlanto-occipital dislocation); **basion-dens interval and basion-axial interval under 12 mm (Harris "rule of 12")**.

2. Craniocervical junction

- **Occipital condyle fractures (Anderson-Montesano):** I comminuted (impaction), II extension of a basilar skull fracture, **III avulsion by the alar ligament** (potentially unstable). Most: collar; occipitocervical fusion if unstable.
- **Atlanto-occipital dissociation:** usually fatal; **children** more often (large head, horizontal condyles). **Traynelis types** (anterior,

longitudinal/distraction, posterior). Detect with **Powers ratio, BDI/BAI**, CT; **never apply traction**; occipitocervical fusion.

3. Atlas (C1) fractures

- **Jefferson (burst) fracture: axial load**, four-part ring fracture (anterior and posterior arches). **Rule of Spence:** combined lateral mass overhang **over 6.9 mm (about 7 mm; 8.1 mm with radiographic magnification)** on the open-mouth view suggests **transverse atlantal ligament (TAL) rupture** (unstable). MRI shows the TAL directly; **Dickman classification of TAL injury** (type I midsubstance rupture: needs **C1-C2 fusion**; type II bony avulsion of the TAL insertion: may heal with immobilisation).
- **Treatment:** stable (intact TAL): **rigid collar/halo**; unstable: **halo** or **C1-C2 (or occipitocervical) fusion**.
- **Neurological injury is rare** (canal widens).

4. Axis (C2) fractures

Odontoid (dens) fractures: Anderson-D'Alonzo (with Grauer modification)

Type	Location	Stability / union	Treatment
I	**Tip avulsion (alar ligament)**, above the transverse ligament	Usually stable (exclude **atlanto-occipital dissociation**)	Collar
II	**Junction of dens and body (waist)**	**Commonest; highest non-union** (watershed blood supply, small contact area)	See below
III	Extends into the **C2 body** (cancellous)	Good union	**Collar or halo**

- **Grauer subtypes of type II: IIA** transverse undisplaced (collar/halo), **IIB** oblique from **anterosuperior to posteroinferior** (or transverse

displaced) = suitable for **anterior odontoid screw** (fracture line perpendicular to the screw), **IIC** oblique from **anteroinferior to posterosuperior** (or comminuted) = **posterior C1-C2 fusion** (screw would shear the fracture).

- **Non-union risk factors for type II: displacement over 5 mm** (also over 4-6 mm), **posterior displacement**, **angulation over 10 degrees**, **age over 50 (about 40-60 in studies)**, smoking, delay in treatment, **comminution**.
- **Elderly (most common cervical fracture over 65):** halo poorly tolerated (aspiration, pneumonia, mortality); options: **rigid collar** (accept stable fibrous union), or **posterior C1-C2 fusion** (Harms: **C1 lateral mass + C2 pars/pedicle screws**; Magerl **C1-C2 transarticular screws**; Gallie/Brooks wiring) for younger elderly patients fit for surgery.
- **Anterior odontoid screw:** needs an **intact TAL**, reducible fracture with a favourable obliquity, and a body habitus allowing the trajectory (**short neck, barrel chest, kyphosis contraindicate it**); osteoporotic bone fails. Preserves C1-C2 rotation (C1-C2 provides about **50% of cervical rotation**).
- **Os odontoideum:** smooth, well-corticated ossicle separated from a hypoplastic dens (probably old unrecognised fracture or developmental); unstable if **sagittal plane translation**: posterior C1-C2 fusion if instability or neurological signs.

Hangman's fracture (traumatic spondylolisthesis of the axis)

- **Bilateral pars (isthmus) fractures of C2**; mechanism **hyperextension with axial load** (diving, car crashes; judicial hanging is hyperextension-distraction). **Neurological injury is uncommon** (canal widens).
- **Levine-Edwards classification:**

Type	Features	Mechanism	Treatment
I	**Minimal translation (under 3 mm; Campbell's: 0-2 mm), no angulation** (vertical fracture line)	Hyperextension + axial load	**Rigid collar**
II	**Over 3 mm translation + significant angulation** (C2-3 disc disrupted)	Hyperextension then flexion	**Reduction (traction in extension) + halo**
IIA	**Severe angulation, minimal translation** (oblique fracture; C2-3 disc and PLL disrupted)	**Flexion-distraction**	**Avoid traction (worsens it)**; reduce with **compression and extension**, halo
III	Pars fracture + **bilateral C2-3 facet dislocation**	Flexion-compression	**Open reduction and fusion** (C2-3 posterior or anterior)

5. Subaxial cervical spine (C3-C7)

Classification

- **SLIC (Subaxial Injury Classification and Severity score, Vaccaro):**

Component	Score
Morphology: no abnormality 0; **compression 1**; burst +1 (=2); **distraction 3**; **rotation/translation 4**	0-4
Disco-ligamentous complex: intact 0; **indeterminate 1**; **disrupted 2**	0-2
Neurology: intact 0; **root injury 1**; **complete cord injury 2**; **incomplete cord injury 3**; **continuous cord compression in the setting of a neurological deficit +1**	0-4

  - **Total under 4: non-operative; 4 = surgeon's discretion; over 4 (5 or more): surgery.**

- **AO Spine subaxial classification:** A (compression), B (tension band injury), C (translation), with facet injury and neurology modifiers.
- **Allen-Ferguson (mechanistic):** compressive flexion, vertical compression, distractive flexion (facet dislocations), compressive extension, distractive extension, lateral flexion.

Specific injuries

- **Facet dislocations (distractive flexion):**
  - **Unilateral facet dislocation:** about **25% anterolisthesis**; rotation; **radiculopathy** commonly; often stable-ish but needs reduction.
  - **Bilateral facet dislocation:** about **50% or more anterolisthesis**; **high rate of cord injury**; very unstable.
  - **Reduction: awake, cooperative patient: closed traction reduction** (Gardner-Wells tongs, incremental weights about 5 lb per level up to about 10 lb/level plus 10 lb for the head, serial neurological examination and radiographs) **is safe without pre-reduction MRI**. **Obtunded/uncooperative patient or failed closed reduction: MRI first** to exclude a **disc herniation** behind the vertebral body (could be pushed into the cord), then open reduction (anterior discectomy first if a disc is present). Then **stabilisation** (anterior cervical discectomy and fusion, or posterior lateral mass fixation).
- **Burst fractures (vertical compression):** retropulsion; anterior corpectomy and fusion if cord compression.
- **Teardrop fracture (flexion-compression): anteroinferior vertebral body fragment** with posterior displacement of the remaining body into the canal and **posterior ligament disruption**: **very unstable**, often **anterior cord syndrome**. Distinguish from the **extension teardrop** (small anteroinferior avulsion of C2 by the ALL in extension: usually stable).

- **Clay-shoveler's fracture:** spinous process avulsion (C7, C6, T1); stable.
- **Ankylosing spondylitis / DISH: extension injuries through the fused spine** ("**chalk-stick**" fractures) are **highly unstable**, frequently missed, often with **epidural haematoma** and late neurological deterioration. **CT/MRI the whole spine; never force into extension** (immobilise in the pre-injury kyphotic position); usually **long posterior fixation** (three levels above and below).
- **Central cord syndrome** (hyperextension in an elderly spondylotic spine, no fracture): see below.
- **Vertebral artery injury:** with **foramen transversarium fractures** and facet dislocations: **CT angiography**; antiplatelet/anticoagulant therapy if no contraindication.

6. Spinal cord injury (SCI)

Assessment

- **ASIA (ISNCSCI) examination: 10 key muscles** (C5 elbow flexors, C6 wrist extensors, C7 elbow extensors, C8 finger flexors (middle finger DIP), T1 little finger abductors; L2 hip flexors, L3 knee extensors, L4 ankle dorsiflexors, L5 long toe extensors, S1 ankle plantarflexors) and **28 sensory dermatomes** (light touch, pin prick), plus **deep anal pressure and voluntary anal contraction (sacral sparing)**.
- **ASIA Impairment Scale:**

Grade	Definition
A	**Complete**: no motor or sensory function in S4-S5
B	Sensory incomplete: sensory but no motor function below the level, including S4-S5
C	Motor incomplete: motor function below the level, **more than half** of key muscles below the level **grade under 3**

Grade	Definition
D	Motor incomplete: **at least half** of key muscles below the level **grade 3 or more**
E	Normal

- **Spinal shock:** temporary loss of all cord function and reflexes below the injury (flaccid paralysis, areflexia), **ends with the return of the bulbocavernosus reflex** (usually within **24-48 hours**). **Completeness can only be judged after spinal shock resolves**: no sacral sparing after the bulbocavernosus reflex returns = complete injury (very poor prognosis for recovery).
- **Neurogenic shock:** loss of sympathetic tone (injuries above **T6**): **hypotension with bradycardia**, warm peripheries. Treat with fluids then **vasopressors** (noradrenaline/phenylephrine), atropine for bradycardia. **Haemorrhagic shock** shows tachycardia and must be excluded first.
- **Maintain mean arterial pressure 85-90 mmHg for about 7 days** (AANS/CNS guideline).
- **Methylprednisolone: not routinely recommended** (NASCIS II/III results not reproducible; complications: infection, GI bleeding). AO Spine 2017 guideline: may be considered within 8 hours (24-hour infusion) as an option; never after 8 hours.
- **Early decompression (within 24 hours)** improves neurological recovery (**STASCIS** study, cervical SCI; AO Spine guideline recommends surgery within 24 hours).

Incomplete cord syndromes

Syndrome	Mechanism	Features	Prognosis
Central cord	**Hyperextension in an elderly spondylotic spine** (often no fracture)	**Upper limbs weaker than lower** (hands worst), sacral sparing, variable sensory loss	**Good**; bowel/bladder and walking usually recover (hands least); **commonest incomplete syndrome**
Anterior cord	Flexion, anterior compression, **anterior spinal artery** injury	**Motor loss and loss of pain/temperature**; **preserved proprioception and vibration (dorsal columns)**	**Worst** prognosis of the incomplete syndromes
Brown-Sequard	Penetrating injury, hemisection	**Ipsilateral motor loss and proprioception loss; contralateral pain and temperature loss** (one to two levels below)	**Best prognosis** (about 90% walk)
Posterior cord	Rare (hyperextension)	Loss of proprioception and vibration only	Good
Conus medullaris	T12-L1 injury	Bowel/bladder/sexual dysfunction, saddle anaesthesia, mixed UMN/LMN; absent bulbocavernosus reflex may persist	Variable
Cauda equina	Below L1	**LMN** pattern: asymmetric lower limb weakness, areflexia, saddle anaesthesia, sphincter dysfunction	Better than conus (peripheral nerves)

Functional levels (complete injuries)

- **C3 and above:** ventilator-dependent (phrenic C3-5).
- **C5:** elbow flexion; power wheelchair (hand control), feeds with adapted devices.
- **C6: wrist extension (tenodesis grip)**; manual wheelchair with adaptations, independent transfers possible.
- **C7: elbow extension (triceps)**: independent transfers and manual wheelchair; **key level for independent living**.
- **C8-T1:** hand function; full wheelchair independence.
- **Upper limb tendon transfers** in tetraplegia: **posterior deltoid or biceps to triceps** (elbow extension), **brachioradialis to ECRB** (wrist extension), **brachioradialis to FPL** (key pinch), nerve transfers.
- **Autonomic dysreflexia** (injuries above T6): noxious stimulus below the injury (**full bladder, faecal impaction**, pressure sores) causes **severe hypertension with bradycardia, headache, flushing/sweating above the lesion**: sit the patient up, remove the stimulus (catheterise), antihypertensives.

7. Paediatric cervical spine

- **Under about 8 years: C1-C3 injuries predominate** (large head, horizontal facets, ligamentous laxity, fulcrum at C2-3); **pseudosubluxation of C2 on C3 (up to about 4 mm)** is normal (**Swischuk line**: posterior arch line C1 to C3 should pass within 2 mm of the C2 spinolaminar line); **ADI up to 5 mm**; **SCIWORA**.
- Use a **backboard with an occipital cut-out or shoulder roll** to avoid neck flexion.
- **Atlantoaxial rotatory subluxation (Fielding-Hawkins types I-IV)**; **Grisel syndrome** (after pharyngeal infection or surgery); **cock-robin** torticollis; dynamic CT. Treat early with collar/traction, halo or fusion for chronic cases.

Exam pearl: **Rule of Spence (lateral mass overhang over about 7 mm) = TAL rupture; Dickman I (midsubstance) needs fusion. Odontoid type II = commonest and highest non-union; IIB = anterior screw, IIC = posterior C1-C2 fusion. Hangman IIA: no traction.**

Exam pearl: **Awake patient with facet dislocation: closed reduction without MRI is safe; obtunded or failed: MRI first. SLIC over 4 = surgery.** Ankylosing spondylitis fractures are highly unstable: CT/MRI whole spine.

Exam pearl: **Central cord (elderly hyperextension, arms worse than legs) is commonest; Brown-Sequard best prognosis; anterior cord worst.** Spinal shock ends with return of the **bulbocavernosus reflex**; neurogenic shock = hypotension + bradycardia (above T6). **C7 = key level for independence.**

Thoracolumbar trauma

1. Anatomy and biomechanics

- The **thoracolumbar junction (T11-L2)** is the commonest site of spinal fractures: transition from the rigid, rib-supported, kyphotic thoracic spine (coronal facets) to the mobile, lordotic lumbar spine (sagittal facets).
- The **thoracic canal is narrow** (and the cord blood supply tenuous: watershed **T4-T9**; **artery of Adamkiewicz** usually left-sided **T9-L2**), so thoracic injuries cause more **complete cord injuries**. The **cord ends (conus) at L1-L2**; below this, injuries affect the **cauda equina** (better prognosis).
- **Denis three-column concept: anterior** (ALL, anterior two-thirds of the body and disc), **middle** (posterior third of body and disc, **PLL**), **posterior** (pedicles, facets, laminae, **posterior ligamentous complex: supraspinous, interspinous ligaments, ligamentum flavum, facet capsules**). **Instability when two or more columns fail** (middle column key in Denis's model).
- The **posterior ligamentous complex (PLC)** is the posterior **tension band**; its integrity is the main determinant of stability in modern classifications. MRI: **STIR hyperintensity** in interspinous region, disrupted black line of the supraspinous ligament/ligamentum flavum; radiographs/CT: **widened interspinous distance**, facet diastasis/perch, **splaying of spinous processes**.

2. Classifications

Denis (1983)

- **Compression** (anterior column only; stable), **burst** (anterior + middle; types A-E), **flexion-distraction (seat-belt/Chance)**, **fracture-dislocation** (all three columns).

TLICS (Thoracolumbar Injury Classification and Severity score, Vaccaro)

Component	Points
Morphology: compression 1, **burst 2**, **translation/rotation 3**, **distraction 4**	1-4
Neurology: intact 0; **nerve root 2**; **complete cord/conus 2**; **incomplete cord/conus 3**; **cauda equina 3**	0-3
PLC: intact 0; **suspected/indeterminate 2**; **injured 3**	0-3

- **Under 4: non-operative; 4: either; over 4 (5 or more): operative.**

AO Spine thoracolumbar classification (2013)

- **Type A (compression):** A0 minor/process fractures; **A1 wedge** (one endplate); **A2 split/pincer** (both endplates, coronal split); **A3 incomplete burst** (one endplate + posterior wall); **A4 complete burst** (both endplates + posterior wall).
- **Type B (tension band failure): B1 transosseous posterior tension band disruption (bony Chance)**; **B2 posterior ligamentous tension band disruption** (with any A injury of the body); **B3 hyperextension** (anterior tension band disruption: ankylosed spines).
- **Type C (displacement/translation in any direction):** all elements fail; **the most unstable**.
- **Neurology modifiers:** N0 intact, N1 transient, N2 radicular, N3 incomplete cord/cauda, N4 complete cord, NX unexaminable; **case modifiers M1 (indeterminate PLC injury), M2 (patient-specific comorbidity: ankylosing spondylitis, osteoporosis, burns)**.
- **TL AOSIS** score guides treatment (similar to TLICS).

3. Specific injuries and their management

Compression (wedge) fractures

- Anterior column only; **stable**. **Non-operative** (brace or early mobilisation as pain allows). Concern if **loss of anterior height over 50%**, **kyphosis over 20-30 degrees**, or multiple contiguous fractures (consider PLC injury).
- **Osteoporotic vertebral compression fractures:** commonest osteoporotic fracture; often asymptomatic; **MRI STIR** shows acute oedema; treat pain, bone health (anti-osteoporotic therapy; **teriparatide** speeds healing). **Vertebroplasty/kyphoplasty** for persistent severe pain (evidence mixed: **sham-controlled trials (Buchbinder, Kallmes) showed no benefit over sham**; some trials (VERTOS IV/ II) differ). **Kümmell disease:** delayed post-traumatic vertebral collapse with an **intravertebral vacuum cleft** (AVN).

Burst fractures

- **Axial load**; anterior + middle column failure with **retropulsion** of the posterior wall into the canal; **interpedicular widening** on AP radiograph.
- **Neurologically intact burst fracture with intact PLC (TLICS 2-3, AO A3/A4 N0): non-operative treatment** (early mobilisation with or without a TLSO brace) gives **similar outcomes to surgery** (**Wood et al.**, RCT: no difference at long-term follow-up; **brace vs no brace (Bailey 2014)**: no difference). **Canal compromise percentage and kyphosis alone do not dictate surgery**; retropulsed fragments remodel (resorption of canal fragments).
- **Surgical indications: neurological deficit with canal compromise**, **PLC disruption**, progressive kyphosis/neurology, **kyphosis over 25-30 degrees** or **height loss over 50%** (relative, debated), polytrauma requiring mobilisation.
- **Surgery: posterior pedicle screw fixation** (short-segment: one level above and below ± screws in the fractured vertebra (**intermediate screws**) improve kyphosis correction; long segment for unstable/osteoporotic), **indirect decompression by ligamentotaxis** (works only if the **PLL is intact** and within about 4 days), **direct**

decompression (posterolateral transpedicular or anterior corpectomy) for significant canal compromise with deficit; **anterior corpectomy and reconstruction** for severe comminution (**load-sharing classification, McCormack: score 7 or more** = anterior column reconstruction needed; points for comminution, apposition, kyphosis correction).

Flexion-distraction (Chance) injuries

- **Lap belt** (axis of flexion anterior to the spine): **tension failure of all three columns**, horizontal fracture through spinous process, lamina, pedicles and body (**bony Chance**, B1) or through ligaments/disc (**ligamentous**, B2).
- **Associated intra-abdominal injuries in about 40-50%** (bowel perforation, mesenteric tears): **"seat-belt sign"** on the abdomen: CT abdomen, surgical review.
- **Treatment: purely bony Chance fracture** can heal in a **hyperextension cast/brace** (if neurologically intact and reduced); **ligamentous or disc injuries do not heal** reliably: **posterior compression instrumentation and fusion**.

Fracture-dislocations (AO C)

- Shear, rotation or translation; **highest rate of neurological injury** (complete cord injury common); **posterior instrumentation (long-segment) with decompression and fusion** in all.

Hyperextension injuries in ankylosed spines (AS, DISH)

- **Long-lever fractures through the fused spine**, highly unstable, frequently missed (**image the whole spine with CT ± MRI**), **epidural haematoma** with delayed neurological deterioration. **Immobilise in the pre-injury (kyphotic) position**; **long posterior fixation (at least 2-3 levels above and below)**. High mortality in the elderly.

4. Sacrum and lumbosacral junction (overlap with pelvis)

- **Spinopelvic dissociation (U/H-type sacral fractures):** falls from height/suicidal jumpers ("**jumper's fracture**"); bilateral vertical sacral fractures with a transverse component; cauda equina injury; **lumbopelvic (triangular) fixation** ± decompression.
- **Roy-Camille** classification of transverse sacral fractures.

5. Neurological injury management (see cervical notes)

- **ASIA examination**; spinal shock (bulbocavernosus reflex); maintain **MAP 85-90 mmHg** for about 7 days; **early decompression (within 24 hours)** where feasible; **no routine steroids**.
- **Conus medullaris syndrome** (T12-L1): mixed UMN and LMN signs, early bowel/bladder involvement, saddle anaesthesia; **cauda equina syndrome** (below L1-L2): LMN pattern, better recovery.
- Rehabilitation: bladder (intermittent catheterisation), bowel programmes, pressure care, DVT prophylaxis (high VTE risk: LMWH within 72 hours), heterotopic ossification (hips), autonomic dysreflexia (above T6), spasticity, psychological care.

6. Penetrating spinal injury

- **Gunshot wounds:** steroids are **contraindicated** (more complications); **surgery rarely improves neurology**; remove bullets only for **progressive deficit**, **CSF leak**, **bullet within the canal at T12-L5 (cauda equina: possible benefit)**, **lead toxicity** risk if in the disc/CSF, or infection; **transcolonic injuries need broad-spectrum antibiotics** (debridement debated).

7. Paediatric thoracolumbar injuries

- **Chance fractures with lap belts** (with abdominal injuries), **apophyseal ring fractures** (slipped vertebral apophysis: adolescent; behaves like a disc herniation; may need excision), **SCIWORA**

(commoner in younger children), **non-accidental injury** (multiple compression fractures). Post-traumatic **scoliosis/kyphosis** after SCI in young children (almost universal before puberty).

8. Complications

- **Post-traumatic kyphosis** (painful; corrective osteotomy: **pedicle subtraction osteotomy** about 30-40 degrees, Smith-Petersen about 10 degrees per level), **non-union/implant failure** (short-segment fixation without anterior support in comminuted fractures), **post-traumatic syrinx** (late neurological deterioration: MRI), **infection**, **dural tears**, **adjacent segment degeneration**, VTE, pressure sores.

Exam pearl: **TLICS: morphology (compression 1, burst 2, translation 3, distraction 4) + neurology + PLC; over 4 = surgery. Neurologically intact burst fracture with intact PLC = non-operative (Wood RCT), with or without a brace.**

Exam pearl: **Chance = lap belt flexion-distraction; 40-50% have abdominal visceral injury; bony Chance can heal in extension cast, ligamentous needs fusion. Ligamentotaxis needs an intact PLL. Load-sharing score 7 or more = anterior reconstruction.**

Exam pearl: **AO B = tension band failure (B1 bony Chance, B2 ligamentous, B3 hyperextension); AO C = translation (most unstable).** Ankylosed spine fractures: whole-spine imaging, long posterior fixation.

Degenerative cervical myelopathy

1. Definitions and pathology

- **Degenerative cervical myelopathy (DCM)** is the umbrella term for cord compression from degenerative disease: **cervical spondylotic myelopathy (CSM)** (disc-osteophyte complexes, ligamentum flavum buckling/hypertrophy, facet hypertrophy, degenerative spondylolisthesis), **OPLL** (ossification of the posterior longitudinal ligament), **ossification of the ligamentum flavum**, and congenital stenosis. **Commonest cause of spinal cord dysfunction in adults over 55.**
- **Static factors:** congenitally narrow canal (**Torg-Pavlov ratio under 0.8**: canal diameter/vertebral body width; **absolute AP canal diameter under 13 mm** = relative stenosis, **under 10 mm** = absolute), disc-osteophyte bars, OPLL.
- **Dynamic factors: extension** narrows the canal (ligamentum flavum buckles; pincer between posteroinferior body and the lamina below); **flexion** stretches the cord over anterior bars; listhesis on flexion-extension films.
- **Ischaemic/cellular injury:** anterior spinal artery territory compromise, demyelination, apoptosis of oligodendrocytes; grey matter (anterior horn) and corticospinal tract involvement.
- **Commonest levels: C5-C6, then C6-C7** (most mobile).

2. Clinical features

- **Symptoms: clumsy hands** (difficulty with buttons, writing, dropping objects), **numb hands** (non-dermatomal), **gait imbalance** (broad-based, spastic, worse in the dark), neck stiffness/pain (not always), Lhermitte sign (electric shock with neck flexion), late **bladder dysfunction**.

- **Upper motor neurone signs:** hyperreflexia, **Hoffmann sign** (flicking the middle finger DIP produces thumb/index flexion), **inverted brachioradialis (radial) reflex** (tapping the brachioradialis gives finger flexion and reduced brachioradialis contraction: lesion at C5-6), **Babinski**, clonus, spasticity, **scapulohumeral reflex** (C3-4 or above), **pectoralis reflex**.
- **"Myelopathy hand" (Ono): finger escape sign** (little and ring fingers drift into abduction/flexion when the fingers are held extended and adducted; ulnar intrinsic weakness), **grip-and-release test** (normal over 20 times in 10 seconds), intrinsic wasting.
- **Gait:** tandem walking difficulty, **Romberg** positive (dorsal column).
- **Radiculopathy may coexist (myeloradiculopathy).**
- **Differentials: motor neurone disease (ALS)** (no sensory signs, fasciculations including the tongue, EMG), **multiple sclerosis** (brain lesions, optic neuritis, oligoclonal bands), **subacute combined degeneration (B12)**, syringomyelia, **normal pressure hydrocephalus** (gait, incontinence, dementia), tumours, peripheral neuropathy, **carpal tunnel syndrome** (may coexist), hereditary spastic paraparesis, copper deficiency, HIV myelopathy, tropical spastic paraparesis (HTLV-1).

3. Severity scoring

- **Modified Japanese Orthopaedic Association (mJOA) score** (18 points: upper limb motor 5, lower limb motor 7, sensory 3, bladder 3): **mild 15-17, moderate 12-14, severe under 12**.
- **Nurick grade (0-5)** based on gait and employment. Others: Cooper, 30-metre walk test, grip and release, Neck Disability Index.

4. Imaging

- **Radiographs:** spondylosis, alignment (**lordosis vs kyphosis**: affects choice of approach), **Torg-Pavlov ratio**, flexion-extension instability, **OPLL** (visible on lateral).

- **MRI (investigation of choice):** cord compression and **intramedullary signal change**:
  - **T2 hyperintensity alone** may be reversible (oedema).
  - **T1 hypointensity with T2 hyperintensity** indicates **myelomalacia/cavitation** (irreversible): **worse prognosis**.
  - "**Snake-eyes**" (bilateral anterior horn T2 hyperintensity on axial images): poor prognosis.
  - Compression ratio, cross-sectional area of the cord.
- **CT: OPLL** extent and type, **double-layer sign** (suggests **dural ossification**: dural tear/CSF leak risk with anterior removal), facet arthropathy, bone quality; **CT myelography** where MRI is contraindicated.
- **OPLL types: continuous, segmental (commonest), mixed, circumscribed (focal)**. Common in **East Asians** (Japan about 2-4%), associated with **DISH**, diabetes, obesity. **Occupying ratio over 60%** and **K-line negative** (OPLL crosses the K-line: line connecting midpoints of the canal at C2 and C7 on lateral) predict **failure of posterior decompression**: use an anterior approach or posterior fusion.
- **Neurophysiology** (SSEPs, MEPs) in uncertain cases and to exclude ALS/neuropathy.

5. Natural history and treatment decisions

- DCM is **usually progressive in a stepwise fashion**; about 20-60% deteriorate without surgery. **Surgery halts progression and often improves function**; outcome depends on **pre-operative severity and duration of symptoms**.
- **AO Spine / WFNS guideline (Fehlings 2017):**
  - **Moderate and severe DCM (mJOA under 15): surgery recommended.**

  - **Mild DCM (mJOA 15-17): surgery or a supervised trial of structured rehabilitation**; operate if it worsens or fails to improve.
  - **Non-myelopathic patients with cord compression on MRI** but radiculopathy (clinical or electrophysiological): higher risk of developing myelopathy: counsel; surgery or close follow-up.
  - **Asymptomatic cord compression without radiculopathy: no prophylactic surgery**; counsel and follow.
- **Predictors of poor outcome: long duration of symptoms**, **severe pre-operative mJOA**, **older age**, **T1 hypointensity/myelomalacia on MRI**, smoking, **multilevel** disease, gait impairment.

6. Surgical options

Anterior approaches (Smith-Robinson; between the carotid sheath laterally and the trachea/oesophagus medially)

- **ACDF (anterior cervical discectomy and fusion):** for **1-2 level disease** with disc-level compression; restores lordosis; direct removal of disc-osteophyte bars.
- **Anterior corpectomy and fusion (ACCF):** compression behind the vertebral body (retrovertebral), focal OPLL, kyphotic multilevel disease. **Hybrid ACDF-ACCF** constructs.
- **Indications for anterior surgery: kyphotic alignment**, **1-3 level** compression, **anterior pathology** (disc, focal OPLL), **K-line negative OPLL**.
- **Complications: dysphagia** (commonest, usually transient; more with multilevel surgery), **recurrent laryngeal nerve palsy** (hoarseness; classically **right-sided** approach (longer variable course of the right RLN) though evidence is mixed), **oesophageal perforation** (rare, serious), **vertebral artery injury** (lateral dissection beyond the uncovertebral joints), **C5 palsy**, **pseudarthrosis** (higher with **multilevel ACDF** and corpectomies),

graft dislodgement (**multilevel corpectomy: add posterior fixation**), **adjacent segment degeneration** (about 2-3% per year, Hilibrand), airway compromise from haematoma (**emergency evacuation** at the bedside), CSF leak (OPLL).

Posterior approaches

- **Laminoplasty** (open-door (**Hirabayashi**), French-door (**Kurokawa**)): **multilevel (3 or more) disease** with **preserved lordosis**; preserves motion; **contraindicated in kyphosis** (cord cannot drift back) and in instability. **Complications: axial neck pain**, **loss of range of motion**, **C5 palsy** (about 5-10%).
- **Laminectomy and fusion (lateral mass screws ± C7/T1 pedicle screws):** multilevel disease with **lordosis or neutral alignment**, instability, axial neck pain; **laminectomy alone is avoided** (**post-laminectomy kyphosis**, especially in young patients and in children).
- **Posterior surgery relies on dorsal drift of the cord**, which needs **lordotic (or at least neutral) alignment** (**cervical lordosis over about 10 degrees**, or **K-line positive** in OPLL).

Approach choice summary

Situation	Preferred approach
1-2 levels, anterior compression	**Anterior (ACDF)**
Kyphosis, focal retrovertebral compression	**Anterior (corpectomy)** or combined
3 or more levels, lordotic, no neck pain	**Laminoplasty**
3 or more levels, lordotic/neutral, instability or neck pain	**Laminectomy + fusion**
OPLL K-line negative / occupying ratio over 60%	**Anterior** or posterior fusion
Multilevel with kyphosis	**Combined anterior-posterior**

C5 palsy

- **Deltoid (± biceps) weakness** appearing **hours to days after surgery** (anterior or posterior), about **5%** (higher after laminoplasty/laminectomy-fusion and corpectomy); proposed mechanisms: **tethering of the C5 root with dorsal cord drift**, foraminal stenosis, reperfusion injury. **Most recover within 6-12 months**; prophylactic **C4-5 foraminotomy** debated.

7. Cervical radiculopathy (related)

- Commonest **C6 (C5-C6 disc)** and **C7 (C6-C7 disc)**. Root = number of the lower vertebra (C5-C6 disc compresses C6 root).

Root	Motor	Reflex	Sensory
C5	Deltoid, biceps	**Biceps**	Lateral arm
C6	**Wrist extensors**, biceps	**Brachioradialis**	Thumb, index, radial forearm
C7	**Triceps**, wrist flexors, finger extensors	**Triceps**	**Middle finger**
C8	Finger flexors	None	Little finger, ulnar forearm
T1	Hand intrinsics	None	Medial arm

- **Spurling test** (extension + rotation + axial compression reproduces radicular pain), **shoulder abduction relief sign** (Bakody).
- **Natural history is favourable**: about 75-90% improve without surgery (analgesia, physiotherapy, selective nerve root injections). **Surgery** for progressive/significant motor deficit or intractable pain after about 6-12 weeks: **ACDF**, **cervical disc arthroplasty** (motion preserving; 1-2 levels without facet arthropathy, instability or OPLL; similar or better outcomes and lower adjacent-level reoperation in

trials), or **posterior foraminotomy** (foraminal soft disc or osteophyte, no instability).

Exam pearl: **Hoffmann sign, inverted radial reflex, finger escape sign, broad-based gait = myelopathy. T1 hypointensity + T2 hyperintensity = myelomalacia (poor prognosis). Moderate/severe DCM (mJOA under 15) = surgery.**

Exam pearl: **Laminoplasty needs lordosis (contraindicated in kyphosis); K-line negative OPLL = anterior. C5 palsy: deltoid weakness after decompression, usually recovers. Dysphagia is the commonest complication of anterior cervical surgery.**

Lumbar disc and stenosis

1. Disc degeneration

- **Intervertebral disc: nucleus pulposus** (type II collagen, high **proteoglycan (aggrecan)** and water content; notochordal origin) and **annulus fibrosus** (type I collagen outer layers, lamellae at about 30 degrees alternating). **Largest avascular structure in the body**: nourished by diffusion through the **vertebral endplates**.
- **Degeneration:** loss of proteoglycans and water, **increased type I : type II collagen ratio**, fissures, **MMPs and cytokines (IL-1, TNF-alpha)**; smoking, genetics (strongest factor in twin studies), age.
- **MRI grading: Pfirrmann (I-V)** for disc degeneration; **Modic endplate changes:**

Modic	T1	T2	Meaning
I	Low	**High**	**Oedema/inflammation**; most associated with **low back pain**; possible low-grade infection (Cutibacterium acnes)
II	**High**	Iso/high	**Fatty marrow replacement**; commonest; stable
III	Low	Low	**Sclerosis**

2. Lumbar disc herniation

Terminology and location

- **Bulge** (over 25% of circumference) vs **herniation**: **protrusion** (base wider than the dome), **extrusion** (dome wider than base, or beyond the disc level), **sequestration** (free fragment).
- **Location: central**, **paracentral (posterolateral, commonest about 90%)**, **foraminal (far lateral, about 5-10%)**, extraforaminal.
- **Root affected:** a **paracentral L4-L5 herniation compresses the traversing L5 root**; a **far lateral (foraminal) L4-L5 herniation**

compresses the exiting L4 root.

- **Commonest levels: L4-L5 and L5-S1 (about 95%)**; peak age 30-50.

Clinical

Root	Motor	Reflex	Sensory
L4	Quadriceps, tibialis anterior	**Knee jerk**	Medial leg/medial malleolus
L5	**EHL** (big toe extension), **gluteus medius** (hip abduction: Trendelenburg), tibialis posterior, ankle dorsiflexors	None reliable (medial hamstring)	**Dorsum of foot, first web space**
S1	Gastrocnemius-soleus (toe walking), peronei, gluteus maximus	**Ankle jerk**	Lateral foot, sole, little toe

- **Straight-leg raise (Lasègue)** positive between 30-70 degrees (L5/S1 roots); **crossed SLR** (pain in the affected leg when lifting the other) is **highly specific**. **Femoral stretch test** for L2-L4 (upper lumbar/foraminal herniations).
- **Red flags:** cauda equina features, progressive motor deficit, fever/weight loss/cancer history (infection/tumour), trauma, age over 50 with new pain, night pain.

Natural history and treatment

- **Favourable:** most improve within 6-12 weeks; **extruded and sequestrated fragments regress most** (resorption; inflammatory response).
- **Non-operative:** activity modification (avoid bed rest), analgesics/NSAIDs, neuropathic agents (limited evidence), physiotherapy, **transforaminal epidural steroid injection** (short-term relief).

- **SPORT trial (Spine Patient Outcomes Research Trial):** both groups improved; **surgery gave faster and greater improvement** in pain and function (as-treated analysis), with high crossover between groups (intention-to-treat inconclusive).
- **Surgical indications: cauda equina syndrome (emergency)**, **progressive or severe motor deficit** (e.g. foot drop: earlier surgery may improve recovery), **intractable radicular pain after about 6 weeks of conservative care** with concordant imaging.
- **Surgery: microdiscectomy** (standard; interlaminar approach), endoscopic discectomy; **far lateral herniation: Wiltse paraspinal (muscle-splitting between multifidus and longissimus) approach** to avoid facetectomy.
- **Complications: recurrent herniation (about 3-7% per Campbell's; up to about 15% in some series; mostly in the first 6 months; more aggressive disc removal does not prevent it)**, **dural tear** (about 1-7%: repair primarily; headache, pseudomeningocele), wrong-level surgery, nerve root injury, **discitis**, rare **vascular injury (anterior annulus perforation: aorta/iliac vessels; hypotension: emergency laparotomy)**.

Cauda equina syndrome (CES)

- **Massive central disc (commonest cause, L4-L5/L5-S1)**, also tumour, haematoma, abscess, trauma.
- **Features: bilateral sciatica**, **saddle anaesthesia** (perianal/perineal), **bladder dysfunction (painless urinary retention with overflow is late: CES-R)**, bowel dysfunction, sexual dysfunction, reduced anal tone.
- **Classification: CES-S** (suspected: bilateral radiculopathy only), **CES-I** (incomplete: altered urinary sensation, reduced desire to void, poor stream, but no retention), **CES-R** (retention with overflow; worse prognosis).

- **Bladder scan: post-void residual over about 200 ml** raises concern (sensitive but not specific).
- **Emergency MRI and decompression as soon as possible** (outcomes better when operated **before retention (CES-I)**; **within 48 hours** classically (Ahn meta-analysis), with current UK/GIRFT guidance for **urgent surgery at the earliest opportunity**).

3. Lumbar spinal stenosis

- **Central stenosis** (ligamentum flavum hypertrophy/buckling, facet hypertrophy, disc bulge, listhesis), **lateral recess** (subarticular; superior articular process hypertrophy compresses the traversing root), **foraminal** (exiting root; disc height loss, osteophytes). **L4-L5 commonest.**
- **Neurogenic claudication:** leg pain/heaviness/paraesthesia with walking and **standing (extension)**, relieved by **sitting or bending forward (flexion)** (**shopping-cart sign**, cycling tolerated better than walking), proximal-to-distal progression, **normal pulses**.
- **Vascular claudication:** pain in the calf with walking, **relieved by standing still** (no need to sit), absent pulses, distal-to-proximal, **cycling also painful**, low ABI.
- **Imaging:** MRI (canal cross-sectional area **under about 75-100 mm²** = stenosis; **AP diameter under 10 mm** = absolute central stenosis; **sedimentation sign**, **Schizas grading** A-D).
- **Treatment:** physiotherapy (flexion-based), analgesia, epidural injections (limited benefit in central stenosis: **LESS trial**: epidural steroid not better than lidocaine alone). **Decompression (laminectomy / laminotomy, undercutting facetectomy preserving over 50% of facets)** for persistent disabling symptoms: **SPORT stenosis cohort** and **Malmivaara** trial showed **surgery better than non-operative care**.
- **Fusion with decompression:** for **degenerative spondylolisthesis with instability**, deformity, or iatrogenic instability. **Two 2016 NEJM**

trials: Försth (Swedish Spinal Stenosis Study): **adding fusion gave no benefit** in stenosis with or without stable spondylolisthesis; **Ghogawala (SLIP)**: fusion gave **slightly better outcomes and fewer reoperations** in **stable grade I degenerative spondylolisthesis**. Result: individualise; fuse when **dynamic instability (over 3-4 mm translation or over 10-15 degrees angulation on flexion-extension)**, high disc height, facet effusion, or sagittal imbalance.

- **Interspinous spacers (X-Stop):** higher reoperation rates than decompression; limited role.

4. Spondylolysis and spondylolisthesis

Wiltse-Newman classification

Type	Name	Features
I	**Dysplastic (congenital)**	Dysplastic L5-S1 facets, **elongated pars**; high risk of progression and **cauda equina** symptoms (intact posterior arch pulls on the roots)
II	**Isthmic**	**Pars defect**: IIA **lytic (stress fracture; commonest in children/adolescents)**, IIB elongated healed pars, IIC acute fracture. **L5-S1 commonest**; **L5 root** compressed in the foramen
III	**Degenerative**	**Women over 50, L4-L5 commonest**, intact pars, facet arthropathy with **sagittal facets**; **L5 root** (lateral recess) and central stenosis
IV	Traumatic	Acute fracture of the posterior elements other than the pars
V	Pathological	Tumour, infection, bone disease
VI	Post-surgical (iatrogenic)	After excessive decompression

- **Marchetti-Bartolozzi classification** (used in Campbell's): **developmental (high or low dysplasia, judged by L5-S1 kyphosis/slip angle and L5/S1 shape)** vs **acquired** (traumatic, post-surgical, pathological, degenerative).
- **Meyerding grade** (percentage slip of the upper vertebra on the lower): **I under 25%, II 25-50%, III 50-75%, IV 75-100%, V over 100% (spondyloptosis)**. **Low grade = I-II; high grade = III or more.**
- **Spinopelvic parameters** (see spinopelvic notes): **high pelvic incidence** predisposes to isthmic slip and progression; **slip angle** (kyphosis at L5-S1) over about 45-50 degrees predicts progression in children.
- **Spondylolysis in adolescents (athletes: gymnasts, cricket fast bowlers, footballers):**
  - Extension-related low back pain; **Scotty dog collar sign** on oblique radiographs; **SPECT/CT, or MRI (STIR oedema) for early stress reaction** (no radiation).
  - Treatment: **activity restriction (about 3 months)** ± brace; most become pain-free even if the defect does not unite. **Direct pars repair** (Buck screw, Scott wiring, pedicle screw-hook) for symptomatic L1-L4 defects/young patients without disc degeneration or slip.
- **Isthmic spondylolisthesis in children: progression is commoner in girls, before or during the growth spurt, with high-grade or dysplastic slips**. **Fusion (L4-S1 or L5-S1 in situ)** for **slip over 50%**, progression, or persistent symptoms. **Phalen-Dickson sign:** knee-flexed, hip-flexed gait with tight hamstrings.
- **Adult isthmic and degenerative spondylolisthesis:** decompression ± fusion (posterolateral, interbody: TLIF/PLIF/ALIF/LLIF). **Reduction of high-grade slips** risks **L5 nerve root palsy** (stretch).

5. Low back pain: key evidence and syndromes

- **Non-specific low back pain:** most resolve; **avoid imaging without red flags**; stay active; NICE: exercise, manual therapy alongside exercise, CBT-based approaches; **avoid opioids/paracetamol alone**; radiofrequency denervation for facet pain in selected patients.
- **Yellow flags:** psychosocial predictors of chronicity (fear-avoidance, depression, compensation).
- **Discogenic pain/fusion:** fusion vs intensive rehabilitation trials (**Fairbank/MRC**: no significant difference; Fritzell (Swedish) showed benefit of fusion). **Disc arthroplasty** for single-level discogenic pain in select patients.
- **Facet joint pain**, **sacroiliac joint dysfunction** (diagnostic blocks), **piriformis syndrome**, **Baastrup disease** (kissing spines), **Bertolotti syndrome** (L5 transverse process pseudoarticulation with sacrum).
- **Failed back surgery syndrome:** wrong diagnosis, wrong level, recurrent herniation, epidural fibrosis, instability, infection, adjacent segment disease; **spinal cord stimulation** for neuropathic leg pain.

> Exam pearl: **Paracentral L4-L5 disc = traversing L5 root; far lateral L4-L5 = exiting L4 root (Wiltse approach). L5 = EHL and first web space; S1 = ankle jerk, lateral foot. Crossed SLR is the most specific sign.**

> Exam pearl: **Cauda equina: saddle anaesthesia, bilateral sciatica, urinary dysfunction: emergency MRI and decompression, ideally before retention (CES-I). SPORT: surgery gives faster and greater improvement for disc herniation and stenosis.**

Exam pearl: **Neurogenic claudication relieved by flexion/sitting (shopping cart, cycling ok); vascular relieved by standing still. Degenerative spondylolisthesis: women over 50, L4-L5, sagittal facets. Isthmic: L5-S1, pars defect, Scotty dog collar.** Meyerding I-V by 25% steps.

Scoliosis and deformity

1. Definitions and classification

- **Scoliosis:** lateral curvature **over 10 degrees (Cobb angle)** with vertebral rotation; a three-dimensional deformity (coronal curve, axial rotation, sagittal **hypokyphosis** in the thoracic spine in idiopathic scoliosis).
- **Types: idiopathic** (about 80%; infantile 0-3, juvenile 4-10, **adolescent 10-skeletal maturity**), **congenital** (vertebral malformation), **neuromuscular** (CP, DMD, SMA, myelomeningocele, Friedreich), **syndromic** (Marfan, NF1, Ehlers-Danlos), **degenerative (de novo adult)**, others (tumour: osteoid osteoma; painful; leg-length discrepancy).
- **Early-onset scoliosis (EOS):** onset **before 10 years** of any cause; risks **thoracic insufficiency syndrome** (impaired lung growth: alveolar development mostly before about 8 years).

2. Adolescent idiopathic scoliosis (AIS)

Epidemiology and features

- **Prevalence about 2-3%** (curves over 10 degrees); **girls:boys about 1:1 for small curves but about 8-10:1 for curves needing treatment**. Most common pattern: **right thoracic** curve (left thoracic curves are atypical: MRI to exclude **syrinx, Chiari malformation, tethered cord**).
- **Painless** (significant pain, neurological signs, **left thoracic curve**, rapid progression, **abnormal abdominal reflexes**, cavus foot, **male with a severe curve**, early onset: get an **MRI of the whole neural axis**).
- **Adams forward bending test: rib hump** (rotation); **scoliometer reading 5-7 degrees or more** = refer for radiograph (screening

threshold about 7 degrees corresponds to about a 20-degree Cobb angle).

Radiographic assessment

- **Standing PA whole spine** (low-dose EOS imaging where available). **Cobb angle:** between the end plates of the **most tilted end vertebrae** (measurement error about ±5 degrees).
- **Risser sign:** ossification of the iliac apophysis (lateral to medial): **0 none, 1 = 25%, 2 = 50%, 3 = 75%, 4 = 100% (not fused), 5 = fused to the ilium**. The **peak height velocity** occurs **before Risser 1** (at about **Risser 0 with open triradiate cartilage**), so Risser underestimates remaining growth. **Sanders skeletal maturity staging** (hand radiograph: digital epiphyses; **Sanders 3 = peak growth**) is a better predictor. **Menarche** usually comes about 1 year after peak height velocity.
- **Lenke classification** (for surgery planning; 6 curve types + lumbar modifier A/B/C + sagittal thoracic modifier −/N/+):

Lenke type	Description	Structural curves
1	**Main thoracic** (commonest)	MT
2	Double thoracic	PT + MT
3	Double major	MT + TL/L
4	Triple major	PT + MT + TL/L
5	Thoracolumbar/lumbar	TL/L
6	Thoracolumbar/lumbar-main thoracic	TL/L (major) + MT

    - A curve is **structural** if it remains **25 degrees or more on side-bending** (or has kyphosis of 20 degrees or more in the PT or TL region). Lumbar modifier: **A** (CSVL between the pedicles of the lumbar apex), **B** (touches apical body/pedicle), **C** (CSVL entirely medial to the apical vertebra).

- **King-Moe classification** (older; types I-V).
- **Other measures:** apical vertebral rotation (**Nash-Moe**), trunk shift, coronal and sagittal balance, **shoulder balance**, rib-vertebra angle difference (**RVAD; Mehta**: in infantile scoliosis, **RVAD over 20 degrees** and **phase 2 rib head overlap** predict progression).

Natural history (progression)

- **Risk of progression rises with: curve magnitude**, **skeletal immaturity** (Risser 0-1, premenarchal, open triradiate, Sanders under 4), **female sex**, **thoracic and double curves**.
- **After maturity:** thoracic curves **over 50 degrees** progress about **1 degree per year**; curves under 30 degrees rarely progress. **Thoracic curves over about 70-100 degrees** reduce pulmonary function (restrictive); **over about 100 degrees** risk cor pulmonale. Lumbar curves cause back pain and degenerative changes later.

Treatment (SRS principles)

Curve (skeletally immature)	Treatment
Under 20-25 degrees	**Observation** (radiographs every 4-6 months)
25-40 (45) degrees, Risser 0-2	**Bracing**
Over 45-50 degrees	**Surgery** (posterior spinal fusion)

- **BrAIST trial (Weinstein, NEJM 2013):** bracing **significantly reduced progression to surgical threshold (50 degrees)** (success about 72% with bracing vs 48% with observation) with a **dose-response**: **wear over about 13 hours/day** gave about 90% success. **Full-time TLSO (Boston)** about 18-23 hours; night-time brace (Charleston, Providence) for single lumbar/TL curves. Bracing **halts progression; it does not correct the curve**. Bracing is ineffective for curves over 45 degrees or after maturity.
- **Scoliosis-specific exercises (Schroth):** some evidence as adjunct.

- **Surgery: posterior spinal instrumentation and fusion with pedicle screws** (all-pedicle-screw constructs give the best three-dimensional correction), selective fusion of structural curves (**Lenke 1: fuse the main thoracic curve only**, lumbar curve corrects spontaneously when lumbar modifier A/B; selective thoracic fusion in some C modifiers). **Anterior fusion** (rarely now) for thoracolumbar/lumbar curves to save levels. **Vertebral body tethering** (growth modulation, in skeletally immature patients; emerging).
- **Neuromonitoring (SSEPs and MEPs; Stagnara wake-up test as backup): MEPs detect motor tract compromise sooner**. On signal loss: raise MAP, check anaesthetic/temperature, release correction, check screws, steroids debated, wake-up test.
- **Complications: neurological deficit (about 0.3-1%)**, infection (late infection with *Cutibacterium acnes*), **pseudarthrosis**, **crankshaft phenomenon** (posterior fusion alone in a very immature spine (Risser 0, open triradiate): anterior growth continues, increasing rotation and curve; prevent with anterior release/fusion or pedicle screws), **adding-on** (curve extends distally), **flatback** (loss of lumbar lordosis with distraction instrumentation: Harrington), **SMA syndrome (superior mesenteric artery syndrome: vomiting after correction from compression of the duodenum)**, **shoulder imbalance**, blood loss (tranexamic acid reduces it), **proximal junctional kyphosis**.

3. Early-onset and infantile scoliosis

- **Infantile idiopathic scoliosis: boys**, **left thoracic** curves, associated plagiocephaly, hip dysplasia; **most resolve**. **Mehta: RVAD over 20 degrees and phase 2 rib head** = progressive. **Mehta casting** (derotational serial casting under GA) can cure early curves.
- **Growth-friendly surgery (EOS): traditional growing rods** (lengthening every 6 months; many operations, autofusion),

magnetically controlled growing rods (MCGR) (non-invasive lengthening), **VEPTR** (vertical expandable prosthetic titanium rib; for thoracic insufficiency, congenital fused ribs), **Shilla** (guided growth). Definitive fusion at maturity. **Law of diminishing returns** with repeated lengthening.

- **Juvenile idiopathic scoliosis:** high progression risk (about 70% need treatment); MRI common (higher rate of neural axis abnormality).

4. Congenital scoliosis

- **Failure of formation** (**hemivertebra**: fully segmented, semi-segmented, non-segmented, incarcerated; wedge vertebra), **failure of segmentation** (**unilateral unsegmented bar**, block vertebra), **mixed**.
- **Worst prognosis: unilateral unsegmented bar with contralateral fully segmented hemivertebra** (progression about **5-10 degrees per year**; **early surgery**). **Block vertebra** has the best prognosis.
- **Associated anomalies (screen everyone): genitourinary (about 20%; about 2-3% obstructive and potentially life-threatening: renal ultrasound)**, **cardiac (about 10-25%: echocardiography)**, **intraspinal (about 20-40%: MRI: diastematomyelia, tethered cord, syrinx)**, **VACTERL**, **Klippel-Feil** (cervical fusion, low hairline, short neck: Sprengel deformity, renal and hearing anomalies), Goldenhar, Alagille.
- **Bracing does not work** (rigid bony curves). **Surgery: hemivertebra excision with short fusion** (posterior-only, at a young age, ideally under about 5), in situ fusion (convex growth arrest/hemi-epiphysiodesis), growth-friendly systems, osteotomies for established curves.
- **Congenital kyphosis** (anterior failure of formation, type I) has a high risk of **paraplegia**: early posterior fusion. Type II (failure of segmentation) is less dangerous.

5. Neuromuscular and syndromic scoliosis

- **Neuromuscular:** long **C-shaped thoracolumbar curves** with **pelvic obliquity**, early onset, progression **after maturity**, poor response to bracing (brace for sitting only). Fuse when sitting balance worsens or curve over about 40-50 degrees; **fusion to the pelvis** if pelvic obliquity (iliac or **S2-alar-iliac screws**); pre-operative pulmonary and nutritional optimisation. **DMD: fuse early (20-30 degrees, FVC over about 30-35%).** Higher complication rates (infection, blood loss, pulmonary).
- **NF1: dystrophic curves** (short, sharp, **rib penciling**, scalloping, dural ectasia, enlarged foramina): **progress rapidly**, need early **anterior and posterior fusion**; non-dystrophic behave like idiopathic. **Modulation** of non-dystrophic into dystrophic curves.
- **Marfan:** larger, more rigid curves, dural ectasia (pedicle screw problems), **higher pseudarthrosis and complication rates**.

6. Kyphosis

- **Normal thoracic kyphosis about 20-45 degrees** (T5-T12); **lumbar lordosis about 40-60 degrees**.
- **Postural kyphosis:** flexible, corrects on prone extension.
- **Scheuermann kyphosis: 3 or more adjacent vertebrae with 5 degrees or more of anterior wedging** (Sorensen), endplate irregularity, **Schmorl nodes**, narrowed discs; **rigid** thoracic kyphosis **over 45 degrees** (does not correct on hyperextension); adolescent males; hamstring tightness, compensatory lumbar hyperlordosis, **spondylolysis association**. **Bracing (Milwaukee/modified)** for curves about 50-75 degrees in skeletally immature patients; **posterior fusion** (with posterior column osteotomies) for **over 75 degrees**, pain or deformity; fuse from the proximal end vertebra to the **sagittal stable vertebra** (to avoid **distal junctional kyphosis**).

- **Congenital kyphosis**, **post-traumatic**, **post-laminectomy** (children, cervical), **tuberculous** (gibbus), **ankylosing spondylitis** (chin-on-chest; **pedicle subtraction osteotomy** at L3 or cervicothoracic osteotomy), **osteoporotic**.

7. Adult deformity (brief; see spinopelvic notes)

- **Adult idiopathic** (untreated AIS) vs **de novo degenerative scoliosis** (lumbar, older patients, asymmetric disc degeneration, rotatory subluxation, stenosis).
- **SRS-Schwab classification:** coronal curve type (T, L, D double, N none) + sagittal modifiers: **PI-LL mismatch (0 under 10 degrees, + 10-20, ++ over 20)**, **pelvic tilt (0 under 20, + 20-30, ++ over 30)**, **SVA (0 under 4 cm, + 4-9.5, ++ over 9.5 cm)**. **Sagittal imbalance correlates most with disability.**
- **Osteotomies: Smith-Petersen/Ponte (posterior column; about 10 degrees per level, needs a mobile anterior disc)**, **pedicle subtraction osteotomy (PSO; about 30-35 degrees, three-column, closing wedge through the vertebra)**, **vertebral column resection (VCR; over 40 degrees, rigid multiplanar deformity; highest neurological risk)**.
- **Complications: proximal junctional kyphosis/failure**, pseudarthrosis (rod fracture at PSO levels), neurological deficit, high blood loss, medical complications.

> Exam pearl: **AIS: girls, right thoracic; left thoracic, pain, neurology or abnormal abdominal reflexes = MRI. Peak growth before Risser 1 (Sanders 3). Treatment: observe under 25, brace 25-40/45 if immature, fuse over 45-50. BrAIST: bracing works, over 13 h/day.**

Exam pearl: **Congenital scoliosis: unilateral unsegmented bar + contralateral hemivertebra = worst; screen kidneys (ultrasound), heart (echo) and spinal cord (MRI). Bracing does not help. Crankshaft: posterior fusion alone in a very immature spine.**

Exam pearl: **Scheuermann: 3 adjacent vertebrae with 5 degrees or more wedging, rigid kyphosis over 45 degrees; fuse to the sagittal stable vertebra. Ponte about 10 degrees/level, PSO about 30 degrees, VCR over 40 degrees.**

Spinal tuberculosis and infection

1. Spinal tuberculosis (Pott's disease)

Epidemiology and pathology

- **Spine is the commonest site of skeletal TB (about 50%)**; India carries the largest TB burden. **Thoracolumbar junction (lower thoracic/upper lumbar) commonest**; **cervical** less common but more dangerous.
- **Haematogenous spread** via **Batson's valveless vertebral venous plexus** or arterial supply.
- **Patterns of vertebral involvement:**
  - **Paradiscal (commonest, about 80%)**: metaphyseal regions on either side of a disc (the disc is destroyed later than in pyogenic infection: **disc relatively preserved early**, then narrowed).
  - **Central (vertebral body)**: concertina collapse, mimics tumour.
  - **Anterior (subperiosteal)**: **anterior scalloping** of several vertebrae beneath the ALL (abscess tracking).
  - **Appendicular/posterior** (posterior elements): less common; early neurological deficit.
- **Cold abscess** tracks along fascial planes: **psoas abscess** (groin/thigh swelling), retropharyngeal (cervical: dysphagia, stridor), paravertebral (**"bird nest" / fusiform paravertebral shadow** on chest radiograph), lumbar triangle (Petit), **skip lesions** in about 5-15% (**MRI the whole spine**).
- **Pathological feature: caseating epithelioid granulomas** with Langhans giant cells.

Clinical features

- Insidious **back pain**, constitutional symptoms (**evening fever, night sweats, weight loss, anorexia**), **stiffness**, **gibbus/knuckle**

deformity (localised kyphosis), cold abscess, **paraplegia (Pott's paraplegia)**.

- **Children:** worse kyphosis progression (growth); **"spine at risk" signs (Rajasekaran)**: **separation of facets**, **posterior retropulsion**, **lateral translation**, **toppling** (tilt of the upper vertebra). **Two or more signs** predict severe progression of kyphosis: consider surgery.

Pott's paraplegia

- **Early-onset (active disease, within about 2 years):** compression by **abscess, granulation tissue, caseous material, sequestra/disc**, or (rarely) **ischaemia/infective thrombosis** of the anterior spinal artery. **Good prognosis** with treatment.
- **Late-onset (healed disease, years later): internal gibbus (bony ridge)**, fibrosis/meningeal scarring, cord atrophy, reactivation. **Poor prognosis**.
- **Tuli's grading** (and **Jain's** modification) of neurological deficit (grades I-IV: negligible to complete paraplegia with sensory/sphincter loss).

Investigations

- **ESR, CRP** raised (non-specific), lymphocytosis; **Mantoux/IGRA** (supportive, not diagnostic in endemic areas).
- **Radiographs:** loss of paradiscal margins, disc narrowing (late), collapse, kyphosis, paravertebral shadow; **changes appear late (about 30-40% bone loss)**.
- **MRI (investigation of choice):** early marrow oedema, **paradiscal involvement with relative disc preservation**, **subligamentous spread across multiple levels**, **large paraspinal/psoas abscess with thin smooth enhancing wall**, skip lesions. (Pyogenic: disc destroyed early, small thick-walled abscess.)
- **CT-guided biopsy** for histology, **AFB smear, culture (liquid MGIT ~2-6 weeks)** and **CBNAAT/GeneXpert MTB/RIF** (rapid; **rifampicin**

resistance; **Xpert Ultra** more sensitive). **Drug susceptibility testing** (MDR-TB rising).

Treatment

- **Antitubercular therapy (India NTEP/WHO): 2 months intensive phase (HRZE: isoniazid, rifampicin, pyrazinamide, ethambutol) + continuation phase HRE (WHO: HR)** for a **total of about 9-12 months for spinal TB** (extended to 18 months by some for slow responders). Pyridoxine with isoniazid. **MDR-TB:** longer regimens (bedaquiline-based, 18-24 months or newer shorter all-oral regimens).
- **Drug toxicity: isoniazid (hepatitis, peripheral neuropathy)**, **rifampicin (hepatitis, orange urine, enzyme induction: oral contraceptives)**, **pyrazinamide (hepatitis, hyperuricaemia/gout)**, **ethambutol (optic neuritis: red-green colour vision)**, streptomycin (ototoxicity, nephrotoxicity).
- **"Middle path regimen" (Tuli): ATT + rest/bracing**; **surgery for specific indications** or failure.
- **Surgical indications:**
  - **Neurological deficit** not improving (or worsening) on ATT, or **severe/rapidly progressive** deficit; **paraplegia in flexion**, **sphincter involvement**, **late-onset paraplegia**.
  - **Instability** and **progressive kyphosis** (children with spine-at-risk signs; **adult kyphosis over about 60 degrees**: risk of late paraplegia).
  - **Large abscesses** not responding (percutaneous drainage often suffices), **diagnostic uncertainty**, **failure of ATT/drug resistance**, **posterior element disease** with compression.
- **Surgery types:**
  - **Anterior radical debridement and strut grafting ("Hong Kong operation", Hodgson)**: historical gold standard.

  - **Posterior or posterolateral (transpedicular/costotransversectomy) debridement with pedicle screw instrumentation and fusion**: now commonly used (single posterior approach can decompress, correct kyphosis and stabilise; titanium implants are safe in TB).
  - **Anterolateral decompression** (Tuli; costotransversectomy, lateral extracavitary).
  - **Deformity correction:** posterior vertebral column resection/closing-opening wedge osteotomies for established kyphosis.
- **Medical Research Council (MRC) trials:** ambulatory chemotherapy with/without bracing was effective; **radical debridement + strut grafting** gave less kyphosis and faster fusion (Hong Kong).

2. Pyogenic spondylodiscitis (vertebral osteomyelitis)

- **Haematogenous** (adults over 50, diabetes, IV drug use, dialysis, immunosuppression, endocarditis) or **post-operative**. **Staphylococcus aureus commonest** (MRSA); gram-negatives (**E. coli** after urinary infection), **Pseudomonas** (IV drug users), streptococci (endocarditis association: **echocardiography** if bacteraemic), *Brucella* (endemic, Middle East/India: farm/raw milk; **Pedro Pons sign** anterosuperior endplate erosion; doxycycline + rifampicin/streptomycin), fungal (immunocompromised).
- **Adults: lumbar commonest**; infection starts in the **subchondral endplate** (segmental arteries) and spreads to the **disc early** (adult disc avascular; children's disc is vascular: primary discitis).
- **Features:** severe back pain, fever (about 50%), raised **CRP/ESR** (CRP used to monitor), **blood cultures positive in about 50-70%**, neurological deficit with **epidural abscess**.
- **MRI: disc T2 hyperintensity and enhancement**, **endplate erosion on both sides of the disc**, marrow oedema, paraspinal/epidural abscess.

- **Diagnosis: blood cultures**; if negative, **CT-guided biopsy before antibiotics** (if the patient is stable and not septic) for culture and histology (TB, fungal, malignancy).
- **Treatment:** targeted IV then oral antibiotics, **total about 6 weeks** (**Bernard et al., Lancet 2015: 6 weeks as effective as 12 weeks** for pyogenic spondylodiscitis), bracing for pain. **Surgery: neurological deficit, epidural abscess with compression**, **instability/deformity**, **failure of medical treatment**, need for diagnosis. **Debridement + instrumentation and fusion** (titanium and even pedicle screws are safe in active infection).

3. Spinal epidural abscess

- **Risk factors:** diabetes, IV drug use, alcohol, immunosuppression, recent spinal procedure, bacteraemia. **S. aureus** (about 60-70%).
- **Classic triad: back pain, fever, neurological deficit** (present in under 20% at presentation). Progression: **back pain, radicular pain, weakness, paralysis**.
- **MRI with gadolinium** (rim-enhancing collection) of the whole spine.
- **Treatment: urgent surgical decompression and drainage + antibiotics** for **neurological deficit** (prognosis depends on pre-operative neurology and duration of paralysis: **paralysis over 24-36 hours rarely recovers**). **Antibiotics alone** only for **neurologically intact** patients without risk factors for failure (close monitoring; **failure predictors: diabetes, MRSA, CRP over 115, WBC over 12,500, positive blood cultures, age over 65**) or patients unfit for surgery/complete paralysis over 72 hours.

4. Post-operative spinal infection

- **Surgical site infection about 1-4%** (higher with instrumentation, posterior approach, long fusions, obesity, diabetes, neuromuscular scoliosis, trauma, prolonged surgery, transfusion).

- **Early (within about 4-6 weeks): debridement, irrigation, retain stable implants**, cultures, IV antibiotics, ± vacuum dressing; **vancomycin powder** in the wound reduces SSI in some studies (prophylaxis).
- **Late/delayed infection:** low-virulence organisms (***Cutibacterium acnes***, coagulase-negative staphylococci); **remove implants if fusion is solid**; keep if not fused and re-instrument as needed.

5. Paediatric discitis

- **Ages 1-5** (vascular channels in the disc), toddlers **refusing to walk, sit or bend**, back or abdominal pain, mild fever; **S. aureus/Kingella**. **MRI**. Treatment: **antibiotics (anti-staphylococcal) and rest/brace**; biopsy rarely needed.

6. Differentiating TB vs pyogenic vs tumour

Feature	TB	Pyogenic	Metastasis/tumour
Onset	**Insidious (months)**	**Acute (days-weeks)**	Gradual, night pain
Disc	**Preserved early, subligamentous spread over several levels**	**Destroyed early**	**Spared**
Abscess	**Large, thin smooth wall, calcification**	Smaller, thick irregular wall	Soft-tissue mass
Posterior elements	Sometimes	Rarely	**Pedicle commonly involved** (winking owl sign)

Feature	TB	Pyogenic	Metastasis/tumour
Skip lesions	Common	Rare	Multiple levels possible
Deformity	**Gibbus/kyphosis**	Less	Pathological collapse

Exam pearl: **Spinal TB: thoracolumbar, paradiscal commonest, disc preserved early, large thin-walled abscess, skip lesions; CBNAAT gives rifampicin resistance.** ATT about 9-12 months; surgery for neurological deficit not improving, instability, progressive kyphosis (children: spine-at-risk signs, 2 or more = surgery).

Exam pearl: **Early-onset Pott's paraplegia (active disease) has a good prognosis; late-onset (healed, internal gibbus) poor. Ethambutol = optic neuritis; INH = neuropathy (give pyridoxine).**

Exam pearl: **Pyogenic spondylodiscitis: S. aureus, disc destroyed early; blood cultures then CT-guided biopsy before antibiotics; 6 weeks of antibiotics (Bernard). Epidural abscess with deficit = urgent decompression.**

Spinal tumours and metastases

1. Overview

- **Metastases are by far the commonest spinal tumours** (about 95% of spinal tumours in adults over 40). The spine is the **commonest site of skeletal metastasis**; **thoracic spine commonest (about 70%)**, then lumbar, cervical.
- **Primary sources (adults): breast, prostate, lung** (together the majority), **kidney, thyroid** (both **hypervascular**: **pre-operative embolisation**), myeloma (haematological). "**BLT with a Kosher Pickle**" (breast, lung, thyroid, kidney, prostate).
- **Spread:** haematogenous (**Batson's plexus**), starting in the **vertebral body (posterior part)**, then **pedicle**; **pedicle destruction ("winking owl" sign)** on AP radiograph is an early radiographic sign. **The disc is usually spared** (vs infection).
- **Location of tumours in the spine:**
  - **Vertebral body:** metastases, myeloma, lymphoma, haemangioma, chordoma, giant cell tumour, Ewing, osteosarcoma.
  - **Posterior elements: osteoid osteoma, osteoblastoma, aneurysmal bone cyst**, osteochondroma.
  - **Intradural extramedullary: meningioma, schwannoma (neurofibroma)** (dumbbell lesions widen foramina).
  - **Intramedullary: ependymoma** (adults), **astrocytoma** (children), haemangioblastoma.

2. Primary benign lesions

Lesion	Typical patient / site	Key features	Treatment
Haemangioma	Commonest benign spinal lesion (about 10% at autopsy), thoracic	**Corduroy/vertical striations** on radiograph, **polka-dot** on axial CT; T1 and T2 bright (fat)	Usually incidental; aggressive symptomatic: embolisation, vertebroplasty, radiotherapy, surgery
Osteoid osteoma	Young males, **posterior elements, lumbar**	**Painful scoliosis** (lesion on the **concave side of the apex**), night pain relieved by **NSAIDs**; nidus under 1.5-2 cm on CT	NSAIDs; **CT-guided radiofrequency ablation** (caution near neural structures) or excision
Osteoblastoma	Young, **posterior elements**, cervical	Nidus over 2 cm, less NSAID-responsive, can be aggressive	**Marginal/en bloc excision** (recurrence)
Aneurysmal bone cyst	Under 20, posterior elements extending into the body	**Fluid-fluid levels** on MRI, expansile	Embolisation, curettage + grafting, sclerotherapy, denosumab (selected)

Lesion	Typical patient / site	Key features	Treatment
Giant cell tumour	20-40, **sacrum** (commonest spinal site), vertebral body	Lytic expansile; can metastasise to lung	En bloc or curettage, **denosumab**
Eosinophilic granuloma (LCH)	**Children**	**Vertebra plana** (flattened body, discs preserved)	Usually observation/brace; heals
Osteochondroma	Posterior elements	HME association, cord compression	Excision if symptomatic

3. Primary malignant lesions

- **Multiple myeloma / solitary plasmacytoma:** commonest primary malignant bone tumour; elderly; **"punched-out" lytic lesions**, **bone scan often cold**; **mini-brain sign** (plasmacytoma). **Radiotherapy** (very radiosensitive) + systemic therapy; surgery for instability/neurological compromise.
- **Chordoma:** notochordal remnant; **sacrococcygeal (about 50%) and clival (about 35%)**, mobile spine less; elderly men; **brachyury**-positive, physaliferous cells; **lytic midline destructive lesion with soft-tissue mass**; **en bloc resection with wide margins** (radioresistant to conventional RT; proton/carbon ion therapy adjuvant); high local recurrence.
- **Chondrosarcoma**, **osteosarcoma**, **Ewing sarcoma** (children; vertebral body), **lymphoma**.
- **Enneking staging and the Weinstein-Boriani-Biagini (WBB) surgical staging system** (12 radiating zones and 5 layers) guide **en**

bloc spondylectomy (Tomita) for primary malignant tumours and solitary metastases in selected cases.

4. Metastatic spine disease: assessment

Presentation

- **Pain** (commonest, about 90%): **mechanical pain** (worse with movement/loading: **instability**), **biological/night pain** (tumour inflammation, responds to steroids/radiotherapy), radicular pain.
- **Metastatic spinal cord compression (MSCC):** an **oncological emergency**; about 5-10% of cancer patients; thoracic commonest. Symptoms: back pain, radicular pain, weakness, sensory level, sphincter disturbance (late). **Whole-spine MRI within 24 hours** of suspicion (**NICE**; immediately if neurological symptoms).
- **Steroids: dexamethasone 16 mg** daily (with PPI) as soon as MSCC is suspected (unless lymphoma is suspected and a biopsy is pending).

Decision frameworks

- **NOMS framework (Memorial Sloan Kettering): N**eurological (degree of epidural spinal cord compression: **Bilsky grade**), **O**ncological (**radiosensitivity** of the tumour), **M**echanical (**instability: SINS**), **S**ystemic (fitness, extent of disease, life expectancy).
- **Bilsky ESCC grade (MRI axial T2): 0** bone only; **1a** epidural impingement, no thecal sac deformation; **1b** deformation without cord abutment; **1c** cord abutment without compression; **2** cord compression with CSF visible; **3** cord compression with no CSF visible. **High-grade (2-3)** compression from radioresistant tumour = surgery (if fit).
- **Radiosensitivity (conventional external beam RT): radiosensitive**: **lymphoma, myeloma, seminoma, breast, prostate** (also small cell lung, Ewing); **radioresistant**: **renal cell carcinoma, melanoma, thyroid, sarcoma, GI (colon), NSCLC**. **Stereotactic body**

radiotherapy (SBRT) overcomes radioresistance in selected cases (**separation surgery** then SBRT: decompress to create a 2-3 mm margin around the cord).

- **SINS (Spinal Instability Neoplastic Score, 0-18):**

Component	Scoring
Location	Junctional (occiput-C2, C7-T2, T11-L1, L5-S1) **3**; mobile spine (C3-C6, L2-L4) **2**; semi-rigid (T3-T10) **1**; rigid (S2-S5) 0
Pain	**Mechanical pain 3**; occasional non-mechanical 1; pain-free 0
Bone lesion	**Lytic 2**; mixed 1; blastic 0
Radiographic alignment	**Subluxation/translation 4**; de novo deformity (kyphosis/scoliosis) 2; normal 0
Vertebral body collapse	Over 50% collapse 3; under 50% 2; no collapse but over 50% body involved 1; none 0
Posterolateral involvement (facet, pedicle, costovertebral joint)	Bilateral 3; unilateral 1; none 0

  - **0-6 stable; 7-12 indeterminate (potentially unstable); 13-18 unstable**: **7 or more = surgical consultation**.

- **Prognostic scores: Tokuhashi revised** (general condition, number of extraspinal bone metastases, number of vertebral metastases, visceral metastases, primary site, paralysis: **score 0-8 = under 6 months**, **9-11 = 6 months or more**, **12-15 = 1 year or more**), **Tomita** (primary tumour growth rate, visceral metastases, bone metastases; guides excisional vs palliative surgery), **SORG**, **NESMS**. **Expected survival over about 3 months** is usually required for surgery.
- **Biopsy** when the primary is unknown (and myeloma/lymphoma excluded first: protein electrophoresis, light chains, CT chest-

abdomen-pelvis, PSA, mammography, bone scan/PET).

5. Treatment of metastatic disease

- **Patchell trial (Lancet 2005):** in MSCC from solid (non-radiosensitive) tumours with a single area of compression, **direct decompressive surgery + radiotherapy was better than radiotherapy alone**: **more patients walked (84% vs 57%) and walked longer**, regained ambulation (62% vs 19%), needed fewer steroids/opioids.
  Exclusions: radiosensitive tumours (lymphoma, myeloma, germ cell), paraplegia over 48 hours, multiple compression levels, life expectancy under 3 months.
- **Radiotherapy alone:** radiosensitive tumours, poor surgical candidates, low-grade compression, stable spine; **single fraction 8 Gy** for pain; 20 Gy/5 fractions common for MSCC (SCORAD trial: single 8 Gy non-inferior for ambulatory status in some groups).
- **Surgery: posterior decompression and instrumented stabilisation** (laminectomy alone without fixation causes instability/kyphosis in vertebral body disease: avoid), **separation surgery + SBRT**, **en bloc spondylectomy** for solitary renal/thyroid metastases with long expected survival, **percutaneous screw fixation** (minimally invasive), **vertebroplasty/kyphoplasty** for painful pathological compression fractures without significant canal compromise (**CAFE trial**: kyphoplasty effective for cancer-related compression fractures).
- **Pre-operative embolisation (within 48 hours of surgery)** for **renal cell, thyroid**, hepatocellular, and other hypervascular tumours (and haemangiomas, ABC, GCT).
- **Bone-targeted agents: bisphosphonates (zoledronic acid)** or **denosumab** reduce skeletal-related events (watch **hypocalcaemia, osteonecrosis of the jaw**).
- **Outcome predictors: pre-treatment ambulatory status (strongest)**, speed of deficit onset (rapid onset worse), primary

tumour type.

6. Intradural tumours (brief)

- **Schwannoma** (commonest intradural extramedullary; **dumbbell** through the foramen; T2 bright, target sign), **meningioma** (thoracic, middle-aged women, dural tail, calcification), **ependymoma** (commonest intramedullary in adults; **myxopapillary ependymoma of the filum terminale/conus**; central, haemosiderin cap), **astrocytoma** (commonest intramedullary in children; eccentric, infiltrative), **haemangioblastoma** (von Hippel-Lindau, cyst with enhancing nodule). MRI with gadolinium; surgical excision under neuromonitoring.

Exam pearl: **Metastasis = commonest spinal tumour; thoracic spine; pedicle destruction (winking owl); disc spared. Renal and thyroid metastases are hypervascular: embolise before surgery. Posterior elements: osteoid osteoma, osteoblastoma, ABC; vertebral body: haemangioma (corduroy), myeloma, chordoma (sacrum/clivus).**

Exam pearl: **SINS 0-6 stable, 7-12 indeterminate, 13-18 unstable (7+ = surgical opinion). NOMS: neurological (Bilsky), oncological (radiosensitivity), mechanical (SINS), systemic. Patchell: surgery + RT beats RT alone for MSCC from solid tumours (84% vs 57% walking).**

Exam pearl: **Radiosensitive: lymphoma, myeloma, breast, prostate, seminoma. Radioresistant: renal, melanoma, thyroid, sarcoma, GI.** Dexamethasone 16 mg and whole-spine MRI within 24 hours for suspected MSCC.

Spinopelvic alignment

1. Why sagittal alignment matters

- **Sagittal malalignment correlates more strongly with pain and disability than coronal deformity** in adult spinal deformity (Glassman, Schwab).
- The spine and pelvis work as a chain to keep the head over the pelvis with minimal muscular effort (**"cone of economy"**, Dubousset). When alignment fails, the body **compensates**: thoracic hypokyphosis, lumbar hyperextension, **pelvic retroversion (increased pelvic tilt)**, **hip extension, knee flexion**, ankle dorsiflexion. Compensation costs energy and causes pain and fatigue.

2. Pelvic parameters (lateral standing radiograph including the femoral heads)

Parameter	Definition	Normal	Notes
Pelvic incidence (PI)	Angle between the line perpendicular to the **S1 endplate at its midpoint** and the line from that midpoint to the **centre of the femoral heads (hip axis)**	About **50-55 degrees** (range about 35-85)	**Fixed morphological parameter** after skeletal maturity (does not change with position); **determines the lumbar lordosis a person needs**
Pelvic tilt (PT)	Angle between the vertical and the line from the hip axis to the S1 endplate midpoint	**Under about 20 degrees** (about 10-20)	**Positional**; **increases (pelvic retroversion) to compensate for sagittal imbalance**

Parameter	Definition	Normal	Notes
Sacral slope (SS)	Angle between the S1 endplate and the horizontal	About 35-45 degrees	Positional; decreases with retroversion

- **Key equation: PI = PT + SS** (geometric).
- **High PI** (large pelvis): needs **more lumbar lordosis**; associated with **isthmic spondylolisthesis** (shear at L5-S1), and with progression of high-grade slips.
- **Low PI**: flatter spine, less lordosis; associated with **Scheuermann**-like flat types and disc degeneration/herniation (Roussouly type 1-2).

3. Spinal parameters

- **Lumbar lordosis (LL):** Cobb angle **L1 to S1** (about 40-60 degrees; **two-thirds of lordosis between L4 and S1**).
- **Thoracic kyphosis (TK):** T4 (or T5) to T12, about **20-50 degrees**.
- **Sagittal vertical axis (SVA):** horizontal distance from the **C7 plumb line** to the **posterosuperior corner of S1**: **normal under about 4-5 cm** (positive = C7 anterior to S1). **Over 5 cm = positive sagittal imbalance** (disability rises steeply).
- **T1 pelvic angle (TPA):** angle between the line from the femoral heads to T1 centroid and the line from the femoral heads to the S1 endplate midpoint; combines SVA and PT and is **independent of posture compensation** (target **under about 14-20 degrees**).
- **Cervical parameters:** C2-C7 lordosis, **C2-C7 SVA (cSVA; under about 4 cm)**, **T1 slope minus cervical lordosis (T1S-CL under about 20 degrees)**, chin-brow vertical angle (CBVA: horizontal gaze; important in ankylosing spondylitis osteotomy planning).
- **Global alignment and proportion (GAP) score** (Yilgor): relative (PI-adjusted) parameters predict mechanical complications after deformity surgery better than absolute targets.

4. Alignment targets (SRS-Schwab)

- **PI-LL mismatch under about 10 degrees** (LL roughly equal to PI ± 10 degrees).
- **PT under about 20 degrees.**
- **SVA under about 5 cm** (under 4 cm in the SRS-Schwab modifier).
- **Age-adjusted targets** (Lafage): older patients tolerate (and need) more positive alignment; **over-correction in the elderly increases proximal junctional kyphosis (PJK)**.

SRS-Schwab sagittal modifier	0	+	++
PI-LL	Under 10 degrees	10-20	Over 20
PT	Under 20 degrees	20-30	Over 30
SVA	Under 4 cm	4-9.5 cm	Over 9.5 cm

- Coronal curve types: **T** (thoracic only, lumbar under 30), **L** (TL/lumbar only), **D** (double), **N** (no major coronal curve).

5. Roussouly classification of normal sagittal shapes

Type	Sacral slope	Lordosis	Notes
1	**Under 35 degrees**	Short, low lordosis; long thoracolumbar kyphosis	Disc degeneration at the TL junction, posterior element impingement at L5-S1
2	**Under 35 degrees**	Flat, long hypolordosis ("flat back")	**Disc herniation, early disc degeneration**
3	**35-45 degrees**	**Balanced, harmonious** (commonest)	

Type	Sacral slope	Lordosis	Notes
4	**Over 45 degrees**	Long, high lordosis	Facet arthritis, **spondylolysis**, posterior element overload
3 anteverted			Variant with low PI and anteverted pelvis

6. Compensation and decompensation

- **Stages:** (1) **thoracic hypokyphosis**, (2) **hyperextension of mobile lumbar segments** (can cause retrolisthesis, facet overload), (3) **pelvic retroversion (PT rises)**: the most important compensatory mechanism, limited by **hip extension reserve** (hip flexion contractures limit compensation), (4) **knee flexion** and ankle dorsiflexion (energy-expensive, "crouched" posture). When exhausted: **SVA becomes positive**, forward stoop, fatigue, inability to stand upright.
- **Flatback syndrome:** loss of lumbar lordosis after **distraction instrumentation (Harrington rods)** into the lumbar spine, after fusions in kyphosis, or degenerative; forward-leaning posture, knee flexion, back pain.
- **Hip-spine syndrome:** coexisting hip and spine pathology; **spinopelvic mobility affects THA**: a **stiff spine (fused or degenerative) with little pelvic tilt change between standing and sitting** raises the risk of **THA dislocation and impingement** (adjust cup anteversion/inclination; consider dual-mobility). **Sitting: pelvis normally tilts posteriorly (retroverts) about 20 degrees, increasing functional cup anteversion.** "Stuck sitting/stuck standing" pelvises.
- **Proximal junctional kyphosis (PJK): proximal junctional angle over 10 degrees and at least 10 degrees greater than pre-**

operatively (Glattes). Risk factors: older age, osteoporosis, **large corrections**, over-correction relative to age-adjusted targets, fusion to the sacrum, **UIV at the thoracolumbar junction**, posterior ligament disruption, high BMI. Proximal junctional failure (fracture, instability, neurological deficit) needs revision. Prevention: teriparatide, tethers, transitional rods, cement augmentation at UIV, soft landing.

7. Corrective osteotomies

Osteotomy	Schwab grade	Correction	Requirement
Smith-Petersen / Ponte (posterior column)	1-2	**About 10 degrees per level** (about 1 degree per mm of resection)	**Mobile anterior column (open disc)**; opening wedge anteriorly
Pedicle subtraction osteotomy (PSO)	3	**About 30-35 degrees** at one level	Fixed (fused) anterior column; three-column **closing wedge** through the vertebra (usually **L3**, below the conus)
Extended PSO / bone-disc-bone	4	About 35-45 degrees	Includes the disc
Vertebral column resection (VCR)	5-6	**Over 40 degrees**, multiplanar	Rigid, severe, angular deformity (congenital kyphoscoliosis, post-TB) ; **highest neurological risk**

- **Ankylosing spondylitis chin-on-chest deformity:** lumbar PSO (L3) for global kyphosis; **cervicothoracic extension osteotomy (C7-T1)** for cervical kyphosis (C7 has a larger canal and the vertebral artery enters at C6); correct to restore horizontal gaze (**CBVA about 0-10 degrees**).

8. Clinical relevance summary

- **Degenerative spondylolisthesis/stenosis surgery:** restore segmental lordosis (especially L4-S1) during fusion; **interbody devices (ALIF/LLIF) give more lordosis than TLIF/PLIF**.
- **Fusing in kyphosis** leads to **adjacent segment disease** and flatback.
- **Lumbar fusion and THA:** fused lumbar spine increases THA dislocation risk; consider dual-mobility cups.
- **Children's isthmic spondylolisthesis:** high PI and high slip angle predict progression; reduction of high-grade slips (restores alignment but **L5 nerve root palsy** risk).

Exam pearl: **PI = PT + SS. PI is fixed (morphological); PT and SS are positional. High PI = more lordosis needed and associated with isthmic spondylolisthesis.**

Exam pearl: **Targets: PI-LL under 10 degrees, PT under 20, SVA under 5 cm (TPA under 20). Pelvic retroversion (high PT) is the key compensation; knee flexion is the last. Sagittal imbalance correlates best with disability.**

Exam pearl: **Ponte/SPO about 10 degrees per level (needs a mobile disc), PSO about 30-35 degrees at one level (usually L3), VCR over 40 degrees (highest neuro risk). Stiff spine + THA = higher dislocation risk.**